JurisFair

——BNPL Credit Monitor

A jurisdiction-aware compliance analytics tool for monitoring fairness, explainability & AI governance in Klarna-style US–EU credit decisions.

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JurisFair
— BNPL Credit Monitor —

One model output. Two regulatory meanings.

Klarna-style BNPL providers operate across legal systems where the same credit-scoring signal can create different compliance obligations. Generic monitoring tools show metrics; they do not translate them into jurisdiction-specific governance action.

UNITED STATES

- ECOA / Regulation B
- Fair lending monitoring
- Adverse-action specificity
- Fair access / debanking review
- Disparity = trigger, not automatic violation

Jurisdiction layer →

EUROPEAN UNION

- EU AI Act — high-risk AI governance
- Bias monitoring & data governance
- Traceability & technical documentation
- Human oversight & post-market monitoring
- Earlier lifecycle intervention

Why AxisReg, why now

AxisReg is a hypothetical growth-stage B2B RegTech provider. The product creates compliance intelligence downstream of underwriting — not a new credit model.

POSITIONING

Generic fairness tools monitor metrics.
JurisFair maps those metrics to jurisdiction-specific compliance actions.

Differentiators

- US/EU configurable rules
- Audit trail and reason mapping
- Model monitoring without underwriting control
- Human-review escalation, not legal automation

BUYERS

Chief Compliance Officer
Model Risk Committee
Legal & Regulatory Affairs

BUSINESS MODEL

B2B SaaS subscription + implementation fee
Optional governance reporting module

GO-TO-MARKET

Start with BNPL providers → expand to digital banks, fintech lenders, and credit platforms in UK & Singapore.

Different political choices create different obligations.

	UNITED STATES	EUROPEAN UNION
Core concern	Fair lending Adverse action Fair banking access	High-risk AI governance Lifecycle risk management
Explanation logic	Accurate, ECOA-specific adverse-action reasons	Traceability · audit trail Tech docs · oversight
Disparity treatment	Monitoring trigger ≠ automatic violation	Early governance intervention ≠ final AI Act determination
Regulatory posture	Fair access + consumer-credit explainability focus	Risk-tiered governance under AI lifecycle controls

Management implication: the tool must separate the numerical signal from the legal meaning attached to that signal.

ARCHITECTURE

Seven layers. One auditable pipeline.

1	2	3	4	5	6	7
Data	**Model**	**Policy**	**Decision**	**Metrics**	**Jurisdiction**	**Human Review**
20k synthetic BNPL applications	Logistic regression → predicted PD	Residual layer → effective PD	effective_pd ≤ cutoff → approved	|RAAD| · AIR FNR gap · PSI	US / EU rules → flags	Compliance, legal model risk teams

Principle

Automation identifies when to look. Humans decide what it means.

The tool sits downstream of credit decision-making. It does not approve or deny consumers. All amber/red outputs require accountable human review.

CORE METRIC

RAAD — Risk-Adjusted Approval Disparity

Question answered:

Do applicants with comparable predicted default risk receive different approval outcomes across audit groups?

|RAAD|

absolute value triggers alerts
Sign retained to identify affected group. Both directions monitored. RAAD is a review trigger — not a legal conclusion.

How it works

- Partition applicants into predicted-PD risk bands
- Per band: approval_rate(ref) − approval_rate(pro)
- Weight gaps by band size
- Alert on |RAAD| versus jurisdiction thresholds

Spreadsheet approval gaps show imbalance. RAAD connects imbalance to comparable risk. The jurisdiction engine converts the number into action.

JURISDICTION CONFIGURATION

EU thresholds are stricter → earlier governance intervention.

Indicator	United States	European Union
|RAAD| Amber	0.05	0.03
|RAAD| Red	0.08	0.05
AIR screening	0.80	0.80
PSI Amber / Red	0.10 / 0.20	0.10 / 0.20
Subgroup PSI	Optional	Required; gap > 0.05 escalates
Reason mapping	Adverse-action specificity	Traceability & auditability
Legal boundary	≠ automatic ECOA violation	≠ final AI Act determination

Same metric layer. Different rule configuration. Different management action.

Approximately the same |RAAD|. Different compliance outcomes.

UNITED STATES

0.042

|RAAD|

AIR 0.694 · PSI 0.000

WATCH

|RAAD| below US amber threshold (0.05), but AIR below 0.80 → continue monitoring and reason-coverage check.

EUROPEAN UNION

0.044

|RAAD|

AIR 0.705 · PSI 0.000

AMBER

|RAAD| exceeds EU amber threshold (0.03) → high-risk AI governance review and oversight log update.

The result changes not because the data changes, but because the jurisdiction configuration changes.

Fixed baseline model tested against nine operating conditions.

Condition	US	EU	Management signal
Baseline	WATCH	AMBER	EU governance review earlier
Economic stress	AMBER	AMBER	Model still needs review
Distribution shift	WATCH	RED	Strongest jurisdiction divergence
Credit tightening	AMBER	RED	Policy layer amplifies disparity
Audit data missing 10/20/30%	AMBER	AMBER	Evidence gap becomes governance issue
Misconfiguration tests	WATCH/AMBER	AMBER	Wrong rules create audit risk

Key finding: distribution shift remains a US monitoring observation but becomes an EU high-risk AI governance escalation.

Failure modes and design response

Failure mode	How it appears	Design response
False positive	Flagged disparity is not substantively meaningful	Confidence intervals · tiered statuses · human review
False negative	Real disparity missed by narrow metrics or missing audit data	|RAAD| + AIR + subgroup drift + missingness escalation
Model drift	Baseline model less reliable under stress or shifted data	PSI monitoring · fixed-model reassessment · subgroup decomposition
Misconfiguration	Wrong jurisdiction rule applied to portfolio	Record data jurisdiction, config applied, rule version · rerun
Over-reliance	Managers treat output as legal judgment	Explicit legal boundary · red/amber require accountable review

BOUNDARIES

What this tool is not

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- Automatic credit approval / denial system
- Final legal advice or legal-violation determination
- Proof that a credit model is fair

✕

- Automatic adverse-action notice generator
- Representation of Klarna's real consumer data
- Replacement for legal, compliance or human review

It does one thing:

Tells compliance teams when to intervene — and preserves the evidence trail for why.

MANAGEMENT ASK

Authorize a controlled pilot

Pilot scope

- One BNPL product line
- Monthly monitoring reports
- Legal-approved US/EU rule configuration
- Post-pilot assessment → UK & Singapore expansion

Success criteria

- Reproducible RAAD / AIR / PSI outputs
- Explainable rule-trigger evidence trail
- No automated legal conclusions
- Human-review workflow accepted by compliance

Commercial pathway

SaaS subscription after pilot + implementation fee + optional governance reporting module.

One model. Two jurisdictions.
One accountable monitoring tool.